

Faculty Development
Program - Day

04

Organised by Groware Education
Solutions, Principal Affiliate of
Kerala State Rutronix



കേരള സ്റ്റേറ്റ് റൂട്രോണിക്സ്





Python List

- List is an ordered sequence of items.
- Most used data type in Python and is very flexible.
- All the items in a list do not need to be of the same type.
- Items separated by commas are enclosed within brackets [].
- To some extent, lists are similar to arrays in C.
- One difference between arrays in C and list is that all the items belonging to a list can be of different data type.

Example

[1, 'Ram', 120, 12.9]



Python List

Example Program

```
first_list = ['abcd', 147, 2.43, 'Tom', 74.9]
small_list = [111, 'Tom']
print(first_list)
print(first_list[0])
print(first_list[-1])
print(first_list[1:3])
print(first_list[2:])
print(small_list * 2)
print(first_list + small_list)
```



Python List – Built-in Functions

- **len(list)** – Gives the total length of the list.

Example Program

```
#Demo of len(list)
```

```
list1 = ['abcd', 147, 2.43, 'Tom']
```

```
print(len(list1))
```



Python List – Built-in Functions

- **max(list)** – Returns item from the list with maximum value.

Example Program

```
#Demo of max(list)
```

```
list1 = [1200, 147, 2.43, 1.12]
```

```
print("Maximum value in: ", list1, "is",  
      max(list1))
```



Python List – Built-in Functions

- **min(list)** – Returns item from the list with minimum value.

Example Program

```
#Demo of min(list)
```

```
list1 = [1200, 147, 2.43, 1.12]
```

```
print("Minimum value in: ", list1, "is",  
      min(list1))
```



Python List – Built-in Methods

- **list.append(obj)** – This method appends an object obj passed to the existing list.

Example Program

```
#Demo of list.append(obj)
list = ['abcd', 147, 2.43, 'Tom']
print("Old List before Append:", list)
list.append(100)
print("New List after Append:", list)
```



Python List – Built-in Methods

- **list.count(obj)** – Returns how many times the object obj appears in a list.

Example Program

```
#Demo of list.count(obj)
```

```
list = ['abcd', 147, 2.43, 'Tom',147,200,147]
```

```
print("The number of times", 147, "appears in",  
      list, "=",list.count(147))
```




Python List – Built-in Methods

- **list.remove(obj)** – Removes object obj from the list.

Example Program

```
#Demo of list.remove(obj)
list1 = ['abcd', 147, 2.43, 'Tom']
list1.remove('Tom')
print(list1)
```



Python List – Built-in Methods

- **list.index(obj)** - Returns index of the object obj if found, otherwise raise an exception indicating that value does not exist.

Example Program

```
#Demo of list.index(obj)  
list1 = ['abcd', 147, 2.43, 'Tom']  
print(list1.index(2.43))
```



Python List – Built-in Methods

- **list.extend(seq)** – Appends the contents in a sequence seq passed to a list.

Example Program

```
#Demo of list.extend(seq)
list1 = ['abcd', 147, 2.43, 'Tom']
list2 = ['def', 100]
list1.extend(list2)
print(list1)
```



Python List – Built-in Methods

- **list.reverse()** – Reverses objects in a list

Example Program

```
#Demo of list.reverse()
```

```
list1 = ['abcd', 147, 2.43, 'Tom']
```

```
list1.reverse()
```

```
print(list1)
```



Python List – Built-in Methods

- **list.insert(index,obj)** – Returns a list with object obj inserted at the given index.

Example Program

```
#Demo of list.insert(index,obj)
list1 = ['abcd', 147, 2.43, 'Tom']
print("List before insertion:",list1)
list1.insert(2,222)
print("List after insertion:",list1)
```



Python List – Built-in Methods

- **list.sort()** – Sorts the items in a list and returns the list. If a function is provided, it will compare using the function provided.

Example Program

```
#Demo of list.sort()
list1 = [1, 147, 2, 43]
print("List before sorting:",list1)
list1.sort()
print("List after sorting:",list1)
```



Python List – Built-in Methods

- **list.pop(obj=list[-1])** – Removes or Returns the last object obj from a list.

Example Program

```
#Demo of list.pop(obj=list[-1])  
list1 = ['abcd', 147, 2.43, 'Tom']  
print("List before popping",list1)  
list1.pop(-1)  
print("List after popping:",list1)
```



Python List – Built-in Methods

- **list.clear()** – Removes all items from a list.

Example Program

```
list1 = ['abcd', 147, 2.43, 'Tom']  
print("List before clearing:",list1)  
list1.clear()  
print("List after clearing:",list1)
```




Python List – Built-in Methods

- **list.copy()** – Returns a copy of the list.

Example Program

```
list1 = ['abcd', 147, 2.43, 'Tom']  
print("List before clearing:",list1)  
list2=list1.copy()  
list1.clear()  
print("List after clearing:",list1)  
print("Copy of the list:",list2)
```